

8.8 DESIGN OF LOW-PASS CHEBYSHEV FILTER

For designing a Chebyshev IIR digital filter, first an analog filter is designed using the given specifications. Then the analog filter transfer function is transformed to digital filter transfer function by using either impulse invariant transformation or bilinear transformation.

The analog Chebyshev filter is designed by approximating the ideal frequency response using an error function. There are two types of Chebyshev approximations. In type-1 approximation, the error function is selected such that the magnitude response is equiripple in the passband and monotonic in the stopband. In type-2 approximation, the error function is selected such that the magnitude function is monotonic in the passband and equiripple in the stopband. The type-2 magnitude response is also called inverse Chebyshev response. The type-1 design is presented in this book.

The magnitude response of type-1 Chebyshev low-pass filter is given by

$$|H_a(\Omega)|^2 = \frac{1}{1 + \varepsilon^2 c_N^2 \left(\frac{\Omega}{\Omega_c} \right)}$$

where ε is attenuation constant given by $\varepsilon = \left[\frac{1}{A_1^2} - 1 \right]^{\frac{1}{2}}$

A_1 is the gain at the passband edge frequency ω_1 and $c_N \left(\frac{\Omega}{\Omega_c} \right)$ is the Chebyshev polynomial

of the first kind of degree N given by

$$\begin{aligned} c_N(x) &= \begin{cases} \cos(N \cos^{-1} x), & \text{for } |x| \leq 1 \\ \cosh(N \cosh^{-1} x), & \text{for } |x| \geq 1 \end{cases} \end{aligned}$$

and Ω_c is the 3 dB cutoff frequency.

The frequency response of Chebyshev filter depends on order N . The approximated response approaches the ideal response as the order N increases. The phase response of the Chebyshev filter is more nonlinear than that of the Butterworth filter for a given filter length N . The magnitude response of type-1 Chebyshev filter is shown in Figure 8.9.

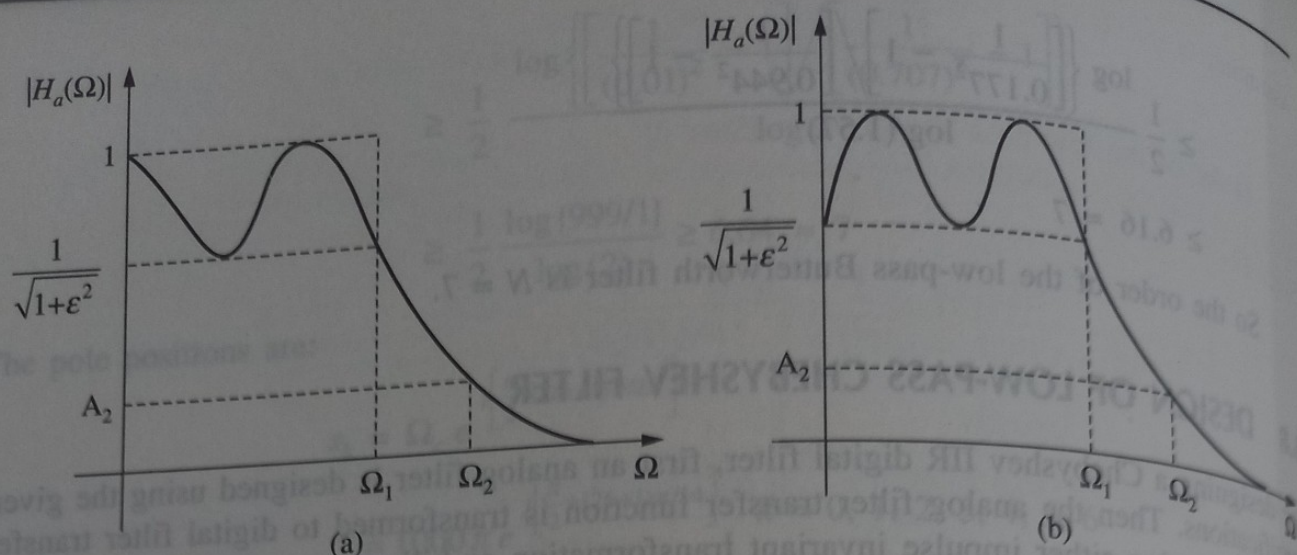


Figure 8.9 Magnitude response of type-1 Chebyshev filter.

The design parameters of the Chebyshev filter are obtained by considering the low-pass filter with the desired specifications as given below.

$$A_1 \leq |H(\omega)| \leq 1 \quad 0 \leq \omega \leq \omega_1$$

$$|H(\omega)| \leq A_2 \quad \omega_2 \leq \omega \leq \pi$$

The corresponding analog magnitude response is to be obtained in the design process.

We have

$$A_1^2 \leq \frac{1}{1 + \varepsilon^2 c_N^2(\Omega_1/\Omega_2)} \leq 1$$

$$\frac{1}{1 + \varepsilon^2 c_N^2(\Omega_1/\Omega_2)} \leq A_2^2$$

Assuming $\Omega_c = \Omega_1$, we will have $c_N(\Omega_1/\Omega_c) = c_N(1) = 1$.

Therefore, from the above inequality involving A_1^2 , we get

$$A_1^2 \leq \frac{1}{1 + \varepsilon^2}$$

Assuming equality in the above equation, the expression for ε is

$$\varepsilon = \left[\frac{1}{A_1^2} - 1 \right]^{\frac{1}{2}}$$

The order of the analog filter, N can be determined from the inequality for A_2^2 .

Assuming $\Omega_c = \Omega_1$,

$$c_N(\Omega_2/\Omega_1) \geq \frac{1}{\varepsilon} \left[\frac{1}{A_2^2} - 1 \right]^{\frac{1}{2}}$$

Since $\Omega_2 > \Omega_1$,

$$\cosh[N \cosh^{-1}(\Omega_2/\Omega_1)] \geq \frac{1}{\varepsilon} \left[\frac{1}{A_2^2} - 1 \right]^{\frac{1}{2}}$$

$$N \geq \frac{\cosh^{-1} \left\{ \frac{1}{\varepsilon} \left[\frac{1}{A_2^2} - 1 \right]^{\frac{1}{2}} \right\}}{\cosh^{-1}(\Omega_2/\Omega_1)}$$

or

Choose N to be the next nearest integer to the value given above. The values of Ω_2 and Ω_1 are determined from ω_1 and ω_2 using either impulse invariant transformation or bilinear transformation.

The transfer function of Chebyshev filters are usually written in the factored form as given below.

$$\text{When } N \text{ is even, } H_a(s) = \prod_{k=1}^{\frac{N}{2}} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2}$$

$$\text{When } N \text{ is odd, } H_a(s) = \frac{B_0 \Omega_c}{s + \Omega_c} \prod_{k=1}^{\frac{N-1}{2}} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2}$$

where

$$b_k = 2y_N \sin\left(\frac{(2k-1)\pi}{2N}\right)$$

$$c_k = y_N^2 + \cos^2\left(\frac{(2k-1)\pi}{2N}\right)$$

$$c_0 = y_N$$

$$y_N = \frac{1}{2} \left\{ \left[\left(\frac{1}{\varepsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\varepsilon} \right]^{\frac{1}{N}} - \left[\left(\frac{1}{\varepsilon^2} + 1 \right)^{\frac{1}{2}} - \frac{1}{\varepsilon} \right]^{\frac{1}{N}} \right\}$$

For even values of N and unity dc gain filter, the parameter B_k are evaluated using the equation:

$$H_a(s) \Big|_{s=0} = \frac{1}{[1 + \varepsilon^2]^{1/2}}$$

For odd values of N and unity dc gain filter, the parameter B_k are evaluated using the equation:

$$H_a(s) \Big|_{s=0} = 1$$

Poles of a normalized Chebyshev filter

The transfer function of the analog system can be obtained from the equation for the magnitude squared response as:

$$H_a(s)H_a(-s) = \frac{1}{1 + \varepsilon^2 c_N^2 \left(\frac{s/j}{\Omega_c} \right)}$$

For the normalized transfer function, let us replace s/Ω_c by s_n .

$$\therefore H_a(s_n)H_a(-s_n) = \frac{1}{1 + \varepsilon^2 c_N^2(-js_n)}$$

The normalized poles in the s -domain can be obtained by equating the denominator of the above equation to zero, i.e., $1 + \varepsilon^2 c_N^2(-js_n)$ to zero.

The solution to the above expression gives us the $2N$ poles of the filter given by

$$s_n = -\sin x \sinh y + j \cos x \cosh y = \sigma_n + j\Omega_n$$

$$\begin{aligned} \text{where } n &= 1, 2, \dots, (N+1)/2 && \text{for } N \text{ odd} \\ &= 1, 2, \dots, N/2 && \text{for } N \text{ even} \end{aligned}$$

and

$$x = \frac{(2n-1)\pi}{2N} \quad n = 1, 2, \dots, N$$

$$y = \pm \frac{1}{N} \sinh^{-1} \left(\frac{1}{\varepsilon} \right) \quad n = 1, 2, \dots, N$$

The unnormalized poles, s'_n can be obtained from the normalized poles as shown below.

$$s'_n = s_n \Omega_c$$

The normalized poles lie on an ellipse in s -plane. Since for a stable filter all the poles should lie in the left half of s -plane, only the N poles on the ellipse which are in the left half of s -plane are considered.

For N even, all the poles are complex and exist in conjugate pairs. For N odd, one pole is real and all other poles are complex and occur in conjugate pairs.

Design procedure for low-pass digital Chebyshev IIR filter

The low-pass Chebyshev IIR digital filter is designed following the steps given below.

Step 1 Choose the type of transformation.
(Bilinear or impulse invariant transformation)

Step 2 Calculate the attenuation constant ϵ .

$$\epsilon = \left[\frac{1}{A_1^2} - 1 \right]^{\frac{1}{2}}$$

Step 3 Calculate the ratio of analog edge frequencies Ω_2/Ω_1 .
For bilinear transformation,

$$\frac{\Omega_2}{\Omega_1} = \frac{\frac{2}{T} \tan \frac{\omega_2}{2}}{\frac{2}{T} \tan \frac{\omega_1}{2}} = \frac{\tan \frac{\omega_2}{2}}{\tan \frac{\omega_1}{2}}$$

For impulse invariant transformation,

$$\frac{\Omega_2}{\Omega_1} = \frac{\omega_2/T}{\omega_1/T} = \frac{\omega_2}{\omega_1}$$

Step 4 Decide the order of the filter N such that

$$N \geq \frac{\cosh^{-1} \left\{ \frac{1}{\epsilon} \left[\frac{1}{A_2^2} - 1 \right] \right\}}{\cosh^{-1} \left\{ \frac{\Omega_2}{\Omega_1} \right\}}$$

Step 5 Calculate the analog cutoff frequency Ω_c .

For bilinear transformation,

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{\frac{2}{T} \tan \frac{\omega_1}{2}}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$$

For impulse invariant transformation

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{\omega_1/T}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$$

Step 6 Determine the analog transfer function $H_a(s)$ of the filter. When the order N is even, $H_a(s)$ is given by

$$H_a(s) = \prod_{k=1}^{N/2} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2}$$

When the order N is odd, $H_a(s)$ is given by

$$H_a(s) = \frac{B_0 \Omega_c}{s + c_0 \Omega_c} \prod_{k=1}^{\frac{N-1}{2}} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2}$$

$$\text{where } b_k = 2y_N \sin \left(\frac{(2k-1)\pi}{2N} \right)$$

$$c_k = y_N^2 + \cos^2 \frac{(2k-1)\pi}{2N}$$

$$c_0 = y_N$$

$$y_N = \frac{1}{2} \left\{ \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\epsilon} \right]^{\frac{1}{N}} - \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\epsilon} \right]^{\frac{-1}{N}} \right\}$$

For even values of N and unity dc gain filter, find B_k 's such that

$$H_a(0) = \frac{1}{(1 + \epsilon^2)^{1/2}}$$

For odd values of N and unity dc gain filter, find B_k 's such that

$$\prod_{k=0}^{\frac{N-1}{2}} \frac{B_k}{c_k} = 1$$

Step 7 (It is normal practice to take $B_0 = B_1 = B_2 = \dots = B_k$) Using the chosen transformation, transform $H_a(s)$ to $H(z)$, where $H(z)$ is the transfer function of the digital filter.

[The high-pass, band pass and band stop filters are obtained from low-pass filter design by frequency transformation].

Properties of Chebyshev filters (Type 1)

1. The magnitude response is equiripple in the passband and monotonic in the stopband.

2. The chebyshev type-1 filters are all pole designs.
3. The normalized magnitude function has a value of $1/\sqrt{1+\epsilon^2}$ at the cutoff frequency Ω_c .
4. The magnitude response approaches the ideal response as the value of N increases.

EXAMPLE 8.26 Design a Chebyshev IIR digital low-pass filter to satisfy the constraints.

$$0.707 \leq |H(\omega)| \leq 1, \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(\omega)| \leq 0.1, \quad 0.5\pi \leq \omega \leq \pi$$

using bilinear transformation and assuming $T = 1$ s.

Solution: Given

$$A_1 = 0.707, \quad \omega_1 = 0.2\pi$$

$$A_2 = 0.1, \quad \omega_2 = 0.5\pi$$

$T = 1$ s and bilinear transformation is to be used. The low-pass Chebyshev IIR digital filter is designed as follows:

Step 1 Type of transformation

Here bilinear transformation is to be used.

Step 2 Attenuation constant ϵ

$$\epsilon = \left[\frac{1}{A_1^2} - 1 \right]^{\frac{1}{2}} = \left[\frac{1}{0.707^2} - 1 \right]^{\frac{1}{2}} = 1$$

Step 3 Ratio of analog edge frequencies, Ω_2/Ω_1 .

Since bilinear transformation is to be used,

$$\frac{\Omega_2}{\Omega_1} = \frac{\frac{2}{T} \tan \frac{\omega_2}{2}}{\frac{2}{T} \tan \frac{\omega_1}{2}} = \frac{\tan \frac{0.5\pi}{2}}{\tan \frac{0.2\pi}{2}} = \frac{2}{0.6498} = 3.0779$$

Step 4 Order of the filter N

$$N \geq \frac{\cosh^{-1} \left\{ \frac{1}{\epsilon} \left[\frac{1}{A_2^2} - 1 \right]^{\frac{1}{2}} \right\}}{\cosh^{-1} \left\{ \frac{\Omega_2}{\Omega_1} \right\}} \geq \frac{\cosh^{-1} \left\{ \frac{1}{1} \left[\frac{1}{0.1^2} - 1 \right]^{\frac{1}{2}} \right\}}{\cosh^{-1} \{3.0779\}} \geq 1.669 \approx 2.$$

Step 5 Analog cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1\right]^{1/2N}} = \frac{\frac{2}{T} \tan \frac{\omega_1}{2}}{\left[\frac{1}{A_1^2} - 1\right]^{1/2N}} = \frac{0.6498}{\left[\frac{1}{0.7077} - 1\right]^{1/4}} = 0.6498$$

Step 6 Analog filter transfer function $H_a(s)$

$$H_a(s) = \prod_{k=1}^{N/2} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2} = \frac{B_1 \Omega_c^2}{s^2 + b_1 \Omega_c s + c_1 \Omega_c^2}$$

$$y_N = \frac{1}{2} \left\{ \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\epsilon} \right]^{\frac{1}{N}} - \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\epsilon} \right]^{\frac{-1}{N}} \right\}$$

$$= \frac{1}{2} \left\{ \left[\left(\frac{1}{1^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{1} \right]^{\frac{1}{2}} - \left[\left(\frac{1}{1^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{1} \right]^{\frac{-1}{2}} \right\}$$

$$= \frac{1}{2} \left\{ [2.414]^{\frac{1}{2}} - [2.414]^{\frac{-1}{2}} \right\} = 0.455$$

$$b_1 = 2y_N \sin \left[\frac{(2k-1)\pi}{2N} \right] = 2 \times 0.455 \sin \left[\frac{(2 \times 1 - 1)\pi}{2 \times 2} \right] = 0.6435$$

$$c_1 = y_N^2 + \cos^2 \left[\frac{(2k-1)\pi}{2N} \right] = (0.455)^2 + \cos^2 \left[\frac{(2 \times 1 - 1)\pi}{2 \times 2} \right] = 0.707$$

For N even,

$$\prod_{k=1}^{\frac{N}{2}} \frac{B_k}{c_k} = \frac{1}{(1 + \epsilon^2)^{0.5}} = 0.707$$

That is $B_1 = c_1 \times 0.707 = 0.707 \times 0.707 = 0.5$.

Therefore, the system function is:

$$H_a(s) = \frac{0.5(0.6498)^2}{s^2 + (0.6435)(0.6498)s + (0.707)(0.6498)^2}$$

On simplifying, we get

$$H_a(s) = \frac{0.2111}{s^2 + 0.4181s + 0.2985}$$

Step 7 Digital filter transfer function $H(z)$

$$\begin{aligned} H(z) &= H_a(s) \bigg|_{s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}} = \frac{0.2111}{s^2 + 0.4181s + 0.2985} \bigg|_{s = 2 \frac{1-z^{-1}}{1+z^{-1}}} \\ &= \frac{0.2111}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 0.4181 \left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right] + 0.2985} \\ &= \frac{0.2111(1+z^{-1})^2}{5.1347 - 7.403z^{-1} + 3.463z^{-2}} \\ &= \frac{0.0411(1+z^{-1})^2}{1 - 1.441z^{-1} + 0.6744z^{-2}} \end{aligned}$$

EXAMPLE 8.27 Determine the system function $H(z)$ of the lowest order Chebyshev IIR digital filter with the following specifications:

3 dB ripple in passband $0 \leq \omega \leq 0.2\pi$

25 dB attenuation in stopband $0.45\pi \leq \omega \leq \pi$

Solution: Given

$$\alpha_1 = 3 \text{ dB}, \quad \therefore k_1 = -3 \text{ dB and hence } A_1 = 10^{k_1/20} = 10^{-3/20} = 0.707$$

$$\alpha_2 = 25 \text{ dB}, \quad \therefore k_2 = -25 \text{ dB and hence } A_2 = 10^{k_2/20} = 10^{-25/20} = 0.0562$$

$$\omega_1 = 0.2\pi \quad \text{and} \quad \omega_2 = 0.45\pi$$

Let $T = 1$ and bilinear transformation is used

$$\text{Attenuation constant } \varepsilon = \left[\frac{1}{A_1^2} - 1 \right]^{\frac{1}{2}} = \left[\frac{1}{0.707^2} - 1 \right] = 1$$

$$\text{Ratio of analog frequencies, } \frac{\Omega_2}{\Omega_1} = \frac{\frac{2}{T} \tan \frac{\omega_2}{2}}{\frac{2}{T} \tan \frac{\omega_1}{2}} = \frac{\tan \frac{0.45\pi}{2}}{\tan \frac{0.2\pi}{2}} = 2.628$$

$$\text{Order of filter } N \geq \frac{\cosh^{-1} \left\{ \frac{1}{\epsilon} \left[\frac{1}{A_2^2} - 1 \right]^{\frac{1}{2}} \right\}}{\cosh^{-1} \left\{ \frac{\Omega_2}{\Omega_1} \right\}}$$

$$\geq \frac{\cosh^{-1} \left\{ \frac{1}{1} \left[\frac{1}{0.0562^2} - 1 \right]^{\frac{1}{2}} \right\}}{\cosh^{-1} \{2.628\}}$$

$$\geq \frac{3.569}{1.621} \geq 2.20 \approx 3$$

$$\text{Analog cutoff frequency } \Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{\frac{2}{T} \tan \frac{\omega_1}{2}}{\left[\frac{1}{0.707^2} - 1 \right]^{1/6}} = 1.708$$

Analog filter transfer function for $N = 3$.

$$H_a(s) = \frac{B_0 \Omega_c}{s + c_0 \Omega_c} \frac{B_1 \Omega_c^2}{s^2 + b_1 \Omega_c s + c_1 \Omega_c^2}$$

$$y_N = \frac{1}{2} \left\{ \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\epsilon} \right]^{\frac{1}{N}} - \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\epsilon} \right]^{\frac{-1}{N}} \right\}$$

$$\therefore y_N = \frac{1}{2} \left\{ \left[\left(\frac{1}{1^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{1} \right]^{\frac{1}{3}} - \left[\left(\frac{1}{1^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{1} \right]^{\frac{-1}{3}} \right\} = 0.5959$$

$$c_0 = y_N = 0.5959$$

$$b_1 = 2y_N \sin \left[\frac{(2 \times 1 - 1) \pi}{2N} \right] = 2 \times 0.5959 \sin \frac{\pi}{6} = 0.5959$$

$$c_1 = y_N^2 + \cos^2 \frac{(2 \times 1 - 1) \pi}{2N} = 0.5959^2 + \cos^2 \frac{\pi}{6} = 1.105$$

$$\text{For } N \text{ odd } \prod_{k=0}^{(N-1)/2} \frac{B_k}{c_k} = 1$$

$$B_0 = c_0 = 0.5959, \quad B_1 = c_1 = 1.105$$

$$\begin{aligned} H_a(s) &= \left(\frac{0.5959 \times 1.708}{s + 0.5959 \times 1.708} \right) \left(\frac{1.105(1.708)^2}{s^2 + 0.5959 \times 1.708s + 1.105(1.708)^2} \right) \\ &= \left(\frac{1.01}{s + 1.01} \right) \left(\frac{3.223}{s^2 + 1.01s + 3.223} \right) \end{aligned}$$

Using bilinear transformation, $H(z)$ is given by

$$\begin{aligned} H(z) &= H_a(s) \Big|_{s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}} = \left(\frac{1.01}{s+1.01} \right) \left(\frac{3.223}{s^2 + 0.1s + 3.223} \right) \Big|_{s = 2 \frac{1-z^{-1}}{1+z^{-1}}} \\ &= \frac{3.25}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 1.01 \right] \left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 0.1 \times 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 3.223 \right]} \\ &= \frac{(3.25)(1+z^{-1})^3}{7.423 - 1.554z^{-1} + 7.023z^{-2}} \end{aligned}$$

EXAMPLE 8.28 The specification of the desired low-pass filter is:

$$0.9 \leq |H(\omega)| \leq 1.0; \quad 0 \leq \omega \leq 0.3\pi$$

$$|H(\omega)| \leq 0.15; \quad 0.5\pi \leq \omega \leq \pi$$

Design a Chebyshev digital filter using the bilinear transformation.

Solution: Given

$$A_1 = 0.9, \quad \omega_1 = 0.3\pi$$

$$A_2 = 0.15, \quad \omega_2 = 0.5\pi$$

The Chebyshev filter is designed as per the following steps:

Step 1 The bilinear transformation is used.

Step 2 Attenuation constant ε

$$\varepsilon = \left[\frac{1}{A_1^2} - 1 \right]^{1/2} = \left[\frac{1}{(0.9)^2} - 1 \right]^{1/2} = 0.484$$

$$= \frac{0.744}{\left(2 \frac{1-z^{-1}}{1+z^{-1}} + 0.577\right) \left(\left(2 \frac{1-z^{-1}}{1+z^{-1}}\right)^2 + 0.577 \times 2 \frac{1-z^{-1}}{1+z^{-1}} + 1.29 \right)}$$

$$= \frac{0.744(1+z^{-1})^3}{(2.577 - 1.423z^{-1})(6.83 - 5.42z^{-1} + 3.75)}$$

EXAMPLE 8.29 Determine the system function of the lowest order Chebyshev digital filter that meets the following specifications.

2 dB ripple in the passband $0 \leq \omega \leq 0.25\pi$

Atleast 50 dB attenuation in stopband $0.4\pi \leq \omega \leq \pi$

Solution: Given

Ripple in passband = 2 dB, i.e. $k_1 = -2$ dB $\therefore A_1 = 10^{k_1/20} = 10^{-2/20} = 0.794$

Attenuation in stopband = 50 dB, i.e. $k_2 = -50$ dB $\therefore A_2 = 10^{k_2/20} = 10^{-50/20} = 0.0031$

$\therefore A_1 = 0.794, \quad \omega_1 = 0.25\pi$

$A_2 = 0.003, \quad \omega_2 = 0.4\pi$

The Chebyshev filter is designed as per the following steps:

Step 1 Type of transformation

Let us choose bilinear transformation.

Step 2 Attenuation constant ε

$$\varepsilon = \left[\frac{1}{A_1^2} - 1 \right]^{1/2} = \left[\frac{1}{0.794^2} - 1 \right]^{1/2} = 0.765$$

Step 3 Ratio of analog edge frequencies, Ω_2/Ω_1

$$\frac{\Omega_2}{\Omega_1} = \frac{\frac{2}{T} \tan \frac{\omega_2}{2}}{\frac{2}{T} \tan \frac{\omega_1}{2}} = \frac{\tan 0.4\pi/2}{\tan 0.25\pi/2} = \frac{1.453}{0.828} = 1.754$$

Step 4 Order of the filter N

$$N \geq \frac{\cosh^{-1} \left[\frac{1}{\varepsilon} \left(\frac{1}{A_2^2} - 1 \right)^{1/2} \right]}{\cosh^{-1} \left(\frac{\Omega_2}{\Omega_1} \right)}$$

$$\begin{aligned} & \cosh^{-1} \left[\frac{1}{0.765} \left(\frac{1}{(0.0031)^2} - 1 \right)^{\frac{1}{2}} \right] \\ & \geq \frac{\cosh^{-1} 1.754}{\cosh^{-1} 1.161} \geq \frac{6.718}{1.161} \geq 5.786 \approx 6 \\ & N = 6 \end{aligned}$$

$\therefore$

Step 5 Analog cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{\frac{1}{2N}}} = \frac{\frac{2}{T} \tan \frac{\omega_1}{2}}{\left[\frac{1}{A_1^2} - 1 \right]^{\frac{1}{2N}}} = \frac{0.828}{\left[\frac{1}{0.794^2} - 1 \right]^{\frac{1}{12}}} = 0.866 \text{ rad/s}$$

Step 6 Analog transfer function $H_a(s)$

$$\text{For } N = 6, H_a(s) = \left(\frac{B_1 \Omega_c^2}{s^2 + b_1 \Omega_c s + c_1 \Omega_c^2} \right) \left(\frac{B_2 \Omega_c^2}{s^2 + b_2 \Omega_c s + c_2 \Omega_c^2} \right) \left(\frac{B_3 \Omega_c^2}{s^2 + b_3 \Omega_c s + c_3 \Omega_c^2} \right)$$

$$y_N = \frac{1}{2} \left\{ \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\epsilon} \right]^{\frac{1}{N}} - \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{\epsilon} \right]^{\frac{-1}{N}} \right\}$$

$$= \frac{1}{2} \left\{ \left[\left(\frac{1}{(0.765)^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{0.765} \right]^{\frac{1}{6}} - \left[\left(\frac{1}{(0.765)^2} + 1 \right)^{\frac{1}{2}} + \frac{1}{0.765} \right]^{\frac{-1}{6}} \right\}$$

$$= \frac{1}{2} \{ 1.197 - 0.83 \} = 0.183$$

$$\therefore c_0 = y_N = 0.183$$

$$c_k = y_N^2 + \cos^2 \frac{(2k-1)\pi}{2N}$$

$$c_1 = y_N^2 + \cos^2 \frac{(2 \times 1 - 1)\pi}{2 \times 6} = (0.183)^2 + \cos^2 \left(\frac{\pi}{12} \right) = 0.9664$$

$$b_1 = 2y_N \sin \frac{(2 \times 1 - 1)\pi}{2 \times 6} = 2 \times 0.183 \sin \left(\frac{\pi}{12} \right) = 0.094$$

$$c_2 = y_N^2 + \cos^2 \frac{(2 \times 2 - 1)\pi}{2 \times 6} = (0.183)^2 + \cos^2 \left(\frac{3\pi}{12} \right) = 0.5334$$

$$b_2 = 2y_N \sin \frac{(2 \times 2 - 1)\pi}{2 \times 6} = 2 \times 0.183 \sin \left(\frac{3\pi}{12} \right) = 0.258$$

$$c_3 = y_N^2 + \cos^2 \frac{(2 \times 3 - 1)\pi}{2 \times 6} = 0.1$$

$$b_3 = 2y_N \sin \frac{(2 \times 3 - 1)\pi}{2 \times 6} = 0.353$$

Let $B_1 = B_2 = B_3$ and let $H_a(0) = 1$.

$$\therefore \frac{B_1 B_2 B_3 \Omega_c^6}{c_1 c_2 c_3 \Omega_c^6} = 1$$

$$\therefore B_1 = B_2 = B_3 = (c_1 c_2 c_3)^{\frac{1}{3}} = (0.964 \times 0.533 \times 0.1)^{\frac{1}{3}} = 0.371$$

$$\begin{aligned} \therefore H_a(s) &= \left(\frac{0.371 \times (0.866)^2}{s^2 + 0.094 \times 0.866s + 0.966 \times (0.866)^2} \right) \\ &\quad \left(\frac{0.371 \times (0.866)^2}{s^2 + 0.258 \times 0.866s + 0.533 \times (0.866)^2} \right) \\ &\quad \left(\frac{0.371 \times (0.866)^2}{s^2 + 0.353 \times 0.866s + 0.1 \times (0.866)^2} \right) \\ &= \left(\frac{0.278}{s^2 + 0.018s + 0.724} \right) \left(\frac{0.278}{s^2 + 0.223s + 0.399} \right) \left(\frac{0.278}{s^2 + 0.305s + 0.074} \right) \end{aligned}$$

Step 7 Digital filter transfer function $H(z)$ taking $T = 1s$.

$$H(z) = H_a(s) \bigg|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \bigg|_{s = 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$\begin{aligned}
 H(z) &= \left[\frac{0.278}{\left\{ 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 0.081 \times 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.724 \right\}} \right] \\
 &\quad \left[\frac{0.278}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 0.223 \left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right] + 0.399 \right]} \right] \\
 &\quad \left[\frac{0.278}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 0.305 \times 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.074 \right]} \right] \\
 &= \left[\frac{0.278(1+z^{-1})^2}{4.886 - 6.552z^{-1} + 4.562z^{-2}} \right] \left[\frac{0.278(1+z^{-1})^2}{4.845 - 7.202z^{-1} + 3.953z^{-2}} \right] \\
 &\quad \left[\frac{0.278(1+z^{-1})^2}{4.684 - 7.852z^{-1} + 3.464z^{-2}} \right]
 \end{aligned}$$